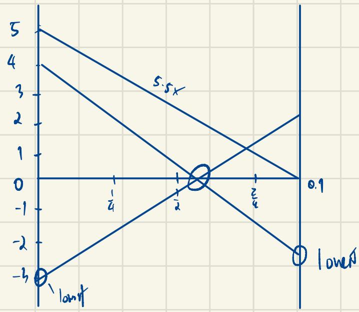
Game Theory

Q. No. 7. Use the reduced game:

$$\begin{bmatrix} 0 & -2 & 2 \\ 5 & 4 & -3 \end{bmatrix}$$

1. Let x be the probability that Player 1 uses Strategy 1.
2. Write the expected payoff against each of Player 2's pure strategies.
3. Plot the three expected payoff lines.
4. Find the optimal mixed strategy for Player 1.
5. Determine Player 2's optimal mixed strategy.
6. Determine Player 2's optimal mixed strategy.



	Prob	Pure strategy	y_1	y_2	y_3
player 1	x	1	0	-2	2
	$1-x$	2	5	4	-3

$E_1 : 0x + 5(1-x)$
$= 5 - 5x$

at $x = 0$: 5
$x = 1$: 0

$E_2 : -2x + 4(1-x)$
$= -2x + 4 - 4x$
$= -6x + 4$

at $x = 0$; 4
$x = 1$; -2

$E_3 = 2x + (-3)(1-x)$
$= 2x - 3 + 3x$
$= 5x - 3$

at $x = 0$; -3
$x = 1$; 2

at Maximin point
$E_2 = E_3$
$-6x + 4 = 5x - 3$
$5x + 6x = 4 + 3$
$11x = 7$
$x = \frac{7}{11}$

at payoff intersection
$E_2 = V = 5x - 3$
$= 5\left(\frac{7}{11}\right) - 3$
$= \frac{35}{11} - 3$
$= \frac{35}{11} - \frac{33}{11}$
$V = \frac{2}{11}$

$\therefore$ player 1 mixed $\left(\frac{7}{11}, \frac{4}{11}, 0\right)$

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player 2

Expect payoff : $y_1(5-5x_1) + y_2(-6x+4) + y_3(5x-3) \leq \frac{2}{11}$

sub $x_1 = \frac{7}{11}$

$: y_1\left(5 - 5\left(\frac{7}{11}\right)\right) + y_2\left(-6\cdot\left(\frac{7}{11}\right) + 4\right) + y_3\left(5\cdot\left(\frac{7}{11}\right) - 3\right) \leq \frac{2}{11}$

$: y_1\left(\frac{20}{11}\right) + y_2\left(\frac{2}{11}\right) + y_3\left(\frac{2}{11}\right) \leq V$

So $y_1 > 0 \rightarrow$ payoff exceed $\frac{2}{11}$, So $y_1 = 0$

So , $: -6xy_2 + 4y_2 + 5xy_3 - 3y_3$

where $y_2 + y_3 = 1 \rightarrow y_2 = 1 - y_3$

$: -6xy_2 + 4y_2 + 5xy_3 - 3y_3$

at $x_1 = 0$: $4y_2 - 3y_3 = \frac{2}{11}$ - (1)

at $x_1 = 1$: $-6y_2 + 4y_2 + 5y_3 - 3y_3 = \frac{2}{11}$

$-2y_2 + 2y_3 = \frac{2}{11}$ - (2)

sub $y_2 = 1 - y_3$ in (1)

$4(1-y_3) - 3y_3 = \frac{2}{11}$

$4 - 4y_3 - 3y_3 = \frac{2}{11}$

$4 - \frac{2}{11} = 7y_3$

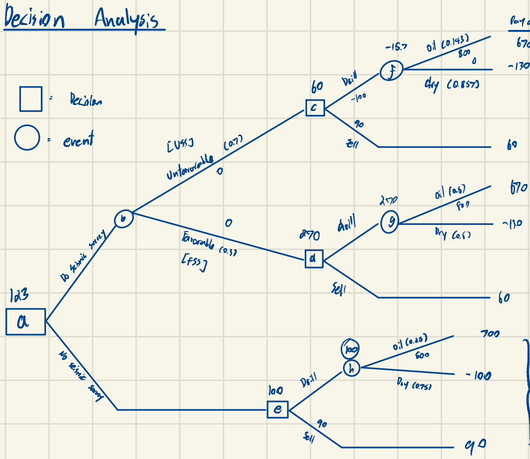
$\frac{42}{11} = 7y_3$

$y_3 = \frac{6}{11}$

$\therefore y_2 = 1 - y_3$
$= 1 - \frac{6}{11}$
$y_2 = \frac{5}{11}$

$\therefore$ player 2 mix $\left(0, \frac{5}{11}, \frac{6}{11}\right)$

Decision Analysis



Drill : 700 (0.25) + (-100)(0.75) = 100

Sell : 90

$P(state_i | Finding_j) = \dfrac{P(Finding_j | state_i) \times P(state_i)}{\sum P(Finding_j | state_k) \times P(state_k)}$

↓

Posterior = $\dfrac{\text{(Likelihood} \times \text{Prior)}}{\text{(Sum of all likelihood} \times \text{Prior)}}$

Alternatives	Oil	Dry
Drill	700	-100
Sell	90	90
Chance	0.25	0.75

Min
-100
90
← Prior

- If oil : 60% FSS, 40% USS
- If Dry : 20% FSS, 80% USS

Oil (0.25) → 0.6 FSS, 0.4 USS

Dry (0.75) → 0.2 FSS, 0.8 USS

Step 1

Find USS

$P(oil | USS) = \dfrac{0.4 \times 0.25}{0.4(0.25) + 0.8(0.75)}$

$= \dfrac{0.1}{0.1 + 0.6} = \dfrac{1}{7}$ ✗

$P(Dry | USS) = 1 - \dfrac{1}{7} = \dfrac{6}{7}$ ✗

Find FSS

$P(Oil | FSS) = \dfrac{0.6 \times 0.25}{0.6(0.25) + 0.2(0.75)}$

$= \dfrac{0.15}{0.15 + 0.15} = \dfrac{1}{2} = 0.5$ ✗

$P(Dry | FSS) = 1 - 0.5 = 0.5$ ✗

Step 2

Expected payoff with Experimentation

- USS $E[\text{Payoff (drill)} | USS] = \frac{1}{7}(700) + \frac{6}{7}(-100) - 30$ ✗

  $= -15.7$

- FSS $E[\text{Payoff (drill)} | FSS] = 0.5(700) + 0.5(-100) - 30 = 270$ ✗

  350 - 50

Step 3

Joint Probabilities   P(state and finding)

Oil & FSS : 0.25 × 0.6 = 0.15

Oil & USS : 0.25 × 0.4 = 0.1

Dry & FSS : 0.75 × 0.2 = 0.15

Dry & USS : 0.75 × 0.8 = 0.6

Step 4

Posterior Probabilities   P(state | finding)

Survey outcome add:
- FSS : 0.15 + 0.15 = 0.3
- USS : 0.1 + 0.6 = 0.7

Total : 1

New Prob: FSS   Oil : $\frac{0.15}{0.3} = \frac{1}{2}$

Dry : $\frac{0.15}{0.3} = \frac{1}{2}$

USS   Oil : $\frac{0.1}{0.7} = \frac{1}{7}$

Dry : $\frac{0.6}{0.7} = \frac{6}{7}$

Step 5   Decision tree

FSS : Drill : $\frac{1}{2} \times 700 + \frac{1}{2} \times (-100) - 30 = 270$ ✗

Sell : 90 - 30 = 60 ✗

USS : Drill = $\frac{1}{7} \times 700 + \frac{6}{7} \times (-100) - 30 = -15.7$ ✗

Sell : 90 - 30 = 60 ✗

Finding	Best choice	Payoff
∴ USS	Sell	60
FSS	Drill	270

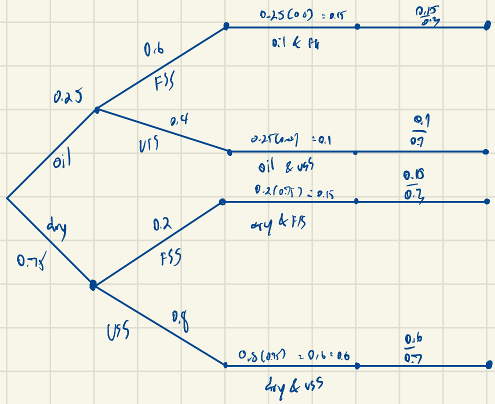
0.15 + 0.15 = 0.3

0.1 + 0.6 = 0.7

Prior

P(state)

Probability tree



Queue theory

The time spent by repairman on his job has an exponential distribution with mean 30 minutes. If the repair sets is approximately Poisson with an average rate of 10 per 8 hours a day, then find the traffic intensity ρ

μ (over "minutes"); λ (under "average rate of 10 per 8 hours")

μ : 30 min , $\frac{1}{30} \times 60 = 2$ set per hour

$\lambda = \frac{10}{8} = \frac{5}{4}$ per hour

traffic intensity $(\rho) = \frac{\lambda}{\mu} = \frac{\frac{5}{4}}{2} = \frac{5}{8}$ ✗

- Customers arrive at a box office with one ticket window according to a poisson's input process with mean rate of 30 per hour. The time required to serve a customer has an exponential distribution with mean 90 second. Then find the traffic intensity (ρ).

λ (next to "mean rate of 30 per hour"); μ (next to "exponential distribution")

บอกจำนวนคน ที่มาภายใน *** ดู

บอกเป็นเวลา "How long until the next patient arrives"

$\lambda = 30$ per hour

$\mu = \frac{90}{3600}$ sec $\rightarrow$ 0.025 hour

if 1 hour $= \frac{1}{0.025} = 40$ customer per hour

$\therefore$ traffic insentive $(\rho) = \frac{\lambda}{\mu} = \frac{30}{40} = \frac{3}{4} \approx 0.75$

$\therefore$ A traffic intensity of 0.75 mean the system is walking at 75% of its capacity. Since $\rho < 1$, the system is stable the queue will not grow overtime. The server is busy 75% of time average.

- In average 96 patients per 24 hours day require the service of an emergency clinic. Also on average, a patient requires 10 minutes of active attention. Then find the traffic intensity (ρ
- **Solution:**

λ (over "96 patients per 24 hours")

$\lambda = \frac{96}{24} = 4$ patient per hour

μ : 10 min $= \frac{10}{60} = \frac{1}{6}$ hour

if 1 hour $\frac{1}{\frac{1}{6}} = 6$ patient per hour

$\therefore$ Traffic intensity $(\rho) = \frac{\lambda}{\mu} = \frac{4}{6} = \frac{2}{3} \approx 0.67$ or 67%

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- λ : Mean arrival rate
- μ : Mean service rate
- L_s : Expected service length
- L_q : Expected queue length
- W_s : Expected waiting time per custom in services
- W_q : Expected waiting time per custom in queue
- L_n : Expected lenght of non-empty queue

Example

94.196 λ

- Patients arrive at the clinic according to Poisson distribution at a rate of 30 patient per hour. The waiting room can not accommodate more than 14 patients. Examination time per patient is exponential with mean rate of 20 per hour. μ

i. Find the effective arrival rate at the clinic

ii. What is the probability that an arriving patient will not wait?

iii. What is the expected waiting time until a patient is discharged from clinic.

1)

$\lambda = 30$

$\mu = 20$

$N = 14$ + 1 pahtient in service

$N = 15$

$\rho = \frac{\lambda}{\mu} = \frac{30}{20} = 1.5$

$P_0 = \frac{1-\rho}{1-\rho^{N+1}} = \frac{1-1.5}{1-1.5^{15+1}} = 0.000762$

$P_N = P_0^{15} = \rho^{15} \times P_0 = 1.5^{15} \times 0.000762 = 0.333$

$\lambda_{eff} = \lambda \times (1-P_N)$

$= 30 \times (1-0.333)$

$\lambda_{eff} = 20$ per hour

$\therefore$ If It have arrival patient 30 person. It will have 10 person can't receive services

2) $P_0 = 0.000762$

3) M/M/1/N queue

$L = \frac{\rho}{1-\rho} - \frac{(N+1)\rho^{N+1}}{1-\rho^{N+1}}$

$= \frac{1.5}{1-1.5} - \frac{(15+1)1.5^{15+1}}{1-1.5^{15+1}}$

$= 13$ patients

Markov

Evaluating the Policy - (Steady-State Analysis)

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- Steady-state equations:
  - $\pi_0 = \pi_3$
  - $\pi_1 = (7/8)\pi_0 + (3/4)\pi_1 + 0\,\pi_2 + 0\,\pi_3$
  - $\pi_2 = (1/16)\pi_0 + (1/8)\pi_1 + (1/2)\pi_2 + 0\,\pi_3$
  - $\pi_3 = (1/16)\pi_0 + (1/8)\pi_1 + (1/2)\pi_2 + 0\,\pi_3$
  - $\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1$
- Solution: $\pi_0 = 2/13$, $\pi_1 = 7/13$, $\pi_2 = 2/13$, $\pi_3 = 2/13$.

$(d_0, d_1, d_2, d_3) = (1, 1, 1, 3)$

$\pi_0 = \pi_3$

$\pi_1 = \frac{7}{8}\pi_0 + \frac{3}{4}\pi_1$

$\pi_1 - \frac{3}{4}\pi_1 = \frac{7}{8}\pi_0$

$\frac{1}{4}\pi_1 = \frac{7}{8}\pi_0$

$\pi_1 = \frac{7}{2}\pi_0$

$\pi_2 = \frac{1}{16}\pi_0 + \frac{1}{8}\pi_1 + \frac{1}{2}\pi_2$

$\frac{1}{2}\pi_2 = \frac{1}{16}\pi_0 + \frac{1}{8}\pi_1$

$\pi_2 = \frac{1}{8}\pi_0 + \frac{1}{4}\pi_1$

$\pi_2 = \frac{1}{8}\pi_0 + \frac{1}{4}\left(\frac{7}{2}\right)\pi_0$

$\pi_2 = \frac{1}{8}\pi_0 + \frac{7}{8}\pi_0$

$\pi_2 = \frac{8}{8}\pi_0$

$\pi_2 = \pi_0$ #

As $\pi_2 = \pi_0 = \pi_3$

we use normalization equation

$\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1$

$\pi_1 = \frac{7}{2}\pi_0$

$\pi_0 + \frac{7}{2}\pi_0 + \pi_0 + \pi_0 = 1$

$\frac{13}{2}\pi_0 = 1$

$\pi_3 = \pi_2 = \pi_0 = \frac{2}{13}$

$1 - \left(\frac{2}{13} + \frac{2}{13} + \frac{2}{13}\right) = \pi_1$

$1 - \frac{6}{13} = \pi_1$

$\pi_1 = \frac{7}{13}$ #

Q.No. 2: Given the costs at each state of the same machine,

- State 0: \$0
- State 1: \$1000
- State 2: \$3000
- State 3: \$6000

Compute the expected long-run average cost: $E(C) = \sum \pi_i C_i$

$E(C) = \sum \pi_i C_i$

$= 0 \cdot \frac{2}{13} + 1{,}000 \cdot \frac{7}{13} + 3000 \cdot \frac{2}{13} + 6000 \cdot \frac{2}{13}$

$E(C) = 1923.08$ per week. #

Simulations

Q.No. 1: Consider the mixed congruential generator:

$$x_{n+1} = (5x_n + 3) \bmod 16$$

with seed $x_0 = 4$.

$$x_{n+1} = (5x_n + 3) \bmod 16$$

a. Generate the first 8 random integers
b. Convert them into uniform random numbers $\rightarrow$ $U = \frac{x+0.5}{M}$, $M=16$
c. Determine the cycle length

$16 \overline{)23}$ = 1, 16, 7 $\qquad 16 \overline{)38}$, 32

$16 \overline{)33}$ = 2, 32, 1

$16 \overline{)80}$ = 0.5

a)

$x_1 = (5\times4+3) \bmod 16 = 23 \bmod 16 = 7$

$x_2 = (5\times7+3) \bmod 16 = 38 \bmod 16 = 6$

$x_3 = (5\times6+3) \bmod 16 = 33 \bmod 16 = 1$

$x_4 = (5\times1+3) \bmod 16 = 8 \bmod 16 = 8$

$x_5 = (5\times8+3) \bmod 16 = 43 \bmod 16 = 11$

$x_6 = (5\times11+3) \bmod 16 = 58 \bmod 16 = 10$

$x_7 = (5\times10+3) \bmod 16 = 53 \bmod 16 = 5$

$x_8 = (5\times5+3) \bmod 16 = 28 \bmod 16 = 12$

b)

$U = \frac{x+0.5}{M}$ where $M=16$

$x_1 = 7 \rightarrow \frac{7+0.5}{16} = 0.46875$

$x_2 = 6 \rightarrow \frac{6+0.5}{16} = 0.40625$

$x_3 = 1 \rightarrow \frac{1+0.5}{16} = 0.09375$

$x_4 = 8 \rightarrow \frac{8+0.5}{16} = 0.53125$

$x_5 = 11 \rightarrow \frac{11+0.5}{16} = 0.71875$

$x_6 = 10 \rightarrow \frac{10+0.5}{16} = 0.65625$

$x_7 = 5 \rightarrow \frac{5+0.5}{16} = 0.34375$

$x_8 = 12 \rightarrow \frac{12+0.5}{16} = 0.78125$

c) Cycle length, it continue sequence until it repeat. If you keep going it eventually return to 4 (seed), so, the cycle length equal 16.

Q.NO. 2: A system has the following probability distribution:

Outcome	**Probability**
A	0.30
B	0.25
C	0.20
D	0.25

a. Construct cumulative probabilities
b. Assign random number intervals (00–99)
c. Using random numbers: 12, 77, 05, 63, 49, 88, 21, 34, determine outcomes

a)

outcome	Probability
A	0.30
B	0.55
C	0.75
D	1.00

b)

outcome	Probability
A	00 – 29
B	30 – 54
C	55 – 74
D	75 – 99

c)

Random num.	
12	A
77	D
5	A
63	C
49	B
88	D
21	A
34	B